

PharmaGuard: Project Brief & Progress Report

Course: B.Tech Major Project (7th Semester) — Department of Information Technology

Institution: Indian Institute of Information Technology, Allahabad (IIIT Allahabad)

Project Group: Group 07

Project Title: PharmaGuard: A Tool-Grounded Agentic AI Architecture for Pharmacovigilance Signal Triage

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Repository & Codebase: <https://github.com/Krishna200608/PharmaGuard> (84/84 Automated Tests Passing)

1. Executive Summary & Problem Statement

1.1 The Real-World Challenge

When a new medicine is approved and prescribed to millions of patients, postmarketing surveillance (**Pharmacovigilance**) continuously monitors for unexpected side effects. Databases like the **FDA Adverse Event Reporting System (FAERS)** receive over 2 million safety reports each year.

Safety teams face a severe **triage bottleneck**: they must rapidly distinguish between:

1. **Real, dangerous safety signals** that need immediate regulatory warnings or label updates (e.g., severe liver damage or cardiac arrest).
2. **Benign statistical noise or co-medication artifacts** (e.g., a diabetic patient on metformin who experiences low blood sugar primarily because they also take insulin).

Currently, this triage step is largely manual, labor-intensive, and prone to delays.

1.2 Why Standard AI (ChatGPT / Raw LLMs) Cannot Be Used Directly

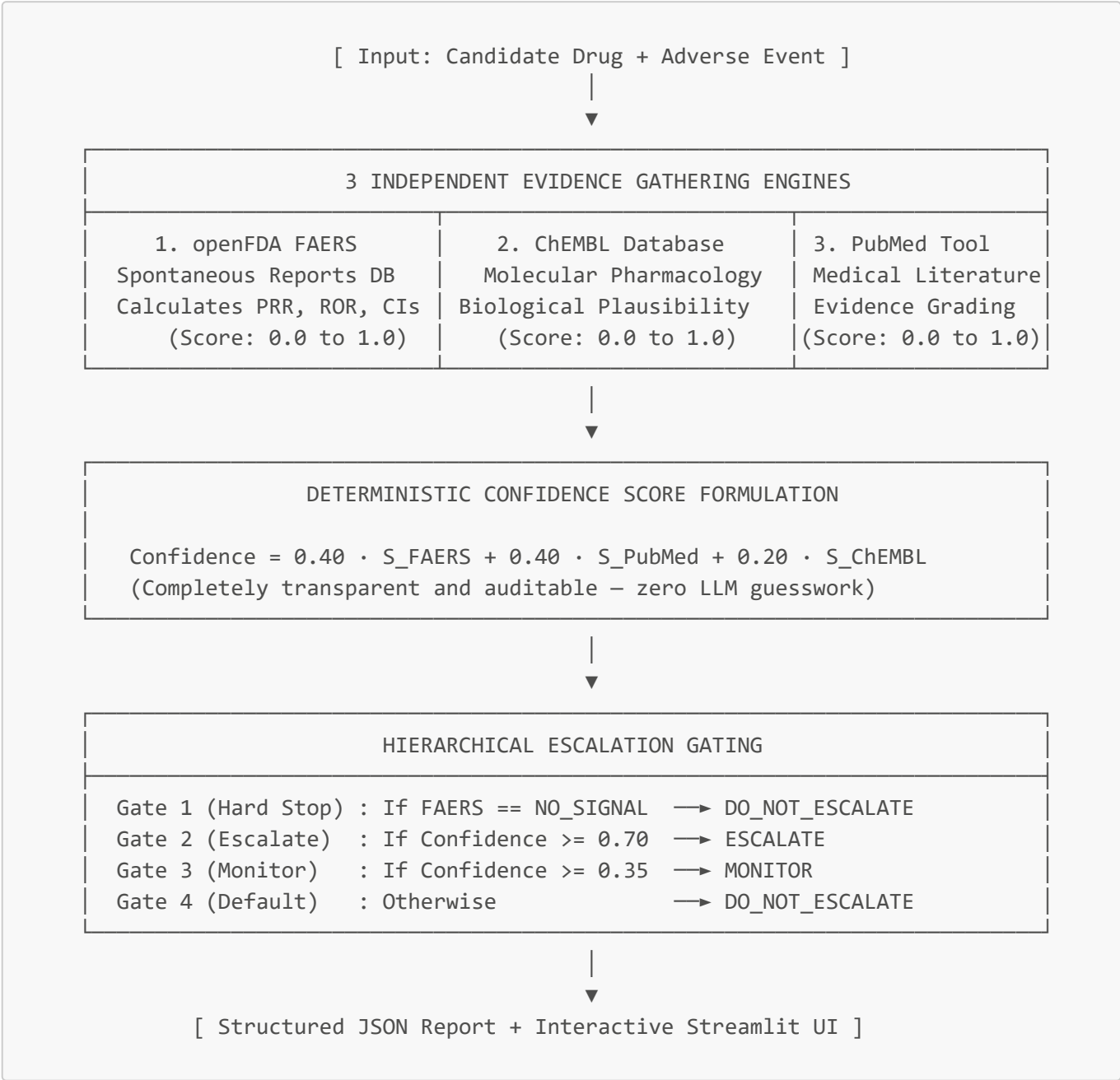
Large Language Models (LLMs) are often proposed to automate clinical triage, but ungrounded LLMs suffer from severe, well-documented flaws:

- **Hallucination & Fabricated Associations:** Recent clinical studies ([Omar et al., Communications Medicine, 2025](#)) found that leading LLMs hallucinate clinical relationships in up to 83% of edge-case scenarios.
- **Temporal Confusion:** An LLM might recall from its training data that a historical safety concern was raised in 2013, but fail to realize that health authorities formally investigated and cleared it in 2014.
- **Uncalibrated Confidence:** A raw LLM cannot provide a mathematically verifiable confidence score—it only provides a self-reported text estimate.

1.3 The PharmaGuard Solution

PharmaGuard is a **tool-grounded, multi-source agentic AI architecture**. Instead of letting the AI answer from memory, PharmaGuard strictly forces the model to fetch and verify real-time data from **three independent biomedical databases** before combining the findings using a **deterministic mathematical formula** to produce an auditable decision: **ESCALATE**, **MONITOR**, or **DO_NOT_ESCALATE**.

2. System Architecture & Methodology



2.1 The Three Evidence Streams

- Real-World Epidemiological Statistics (openFDA FAERS):** Queries the live FDA database to construct 2×2 contingency tables and compute the **Proportional Reporting Ratio (PRR)** and **Reporting Odds Ratio (ROR)** with exact log-normal 95% confidence intervals (Evans et al., 2001; van Puijenbroek et al., 2002).
- Molecular Pharmacology (ChEMBL Target Database):** Checks whether the drug’s biological mechanism of action (MoA) makes it biologically plausible that it could cause the symptom (Gaulton et al., Nucleic Acids Res, 2019). Ratings: **HIGH** (1.0), **MODERATE** (0.5), **LOW** (0.0), **UNKNOWN** (0.0).

- 3. **Biomedical Literature Grading (NCBI PubMed E-Utilities):** Searches published medical literature via NCBI E-utilities and grades evidence under a strict rubric: **Grade A** (1.0) for statistically significant clinical findings, **Grade B** (0.5) for case reports/series, and **Grade C** (0.0) for unconfirmed or negative literature.

2.2 The Deterministic Safety Gate

If a drug–event pair has **no statistical signal in FAERS** ($\$S_{\{\text{FAERS}\}} = \text{NO_SIGNAL}\$$), **Gate 1 unconditionally outputs DO_NOT_ESCALATE**. This prevents the system from generating false alarms based purely on theoretical literature discussions.

3. Work Completed So Far

- 1. **Core Pipeline Implementation:** Full end-to-end Python pipeline with dual orchestration modes:
 - **Fixed Pipeline Agent (pharmaguard/agent/fixed_pipeline.py):** Deterministic sequence (FAERS $\$ \rightarrow$ ChEMBL $\$ \rightarrow$ PubMed $\$ \rightarrow$ Combine).
 - **ReAct LangGraph Agent (pharmaguard/agent/react_agent.py):** LLM-directed dynamic tool-use graph (Yao et al., ICLR, 2023).
- 2. **Disk-Backed Tool Cache (pharmaguard/tools/cache.py):** Schema-versioned persistent cache (diskcache, v7), enabling 100% zero-cost, offline, reproducible evaluation.
- 3. **Epistemic Auditing & Confounding Modules:**
 - **Adversarial Mechanistic Leakage Critic (pharmaguard/tools/chembl_tool.py):** A secondary blind critic (inspired by the MARCH framework, ACL 2026) that detects if the primary agent leaked pre-trained regulatory memory.
 - **Confounding-Aware Discounting Tool (pharmaguard/tools/confounding.py):** Automatically discounts confounded signals caused by multi-drug regimens (e.g. polypharmacy in diabetic populations).
- 4. **Curated Benchmark Datasets:**
 - **Core Benchmark (15 pairs):** 7 Confirmed Positives (FDA Boxed Warnings), 5 Genuine Negative Controls, 3 Zero-Report Edge Cases with full regulatory citations.
 - **Secondary OMOP Pilot Benchmark (32 pairs):** Derived from the international OHDSI OMOP reference set (Ryan et al., Drug Safety, 2013) across 4 clinical endpoints.
- 5. **Interactive Streamlit Evaluation Dashboard (scripts/dashboard.py):**
 - Complete 6-view web interface with dynamic Light/Dark theme switching, high-resolution metric cards, dense comparison tables, and Plotly confidence waterfall charts.
- 6. **Automated Verification Suite:**
 - 84/84 unit and integration tests passing (pytest).
 - Automated 1080p screenshot suite via Playwright (scripts/dev/capture_screenshots.py).

4. Experimental Results & Benchmarks

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SUMMARY OF EXPERIMENTAL RESULTS
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Core Benchmark (15 pairs):	Strict F1: 0.923 (Precision=1.000, Recall=0.857) Lenient F1: 0.933 (Precision=0.875, Recall=1.000) Specificity: 1.000 (8/8 negative controls cleared) Spurious False Alarms: FP = 0
Single-Shot LLM Baseline: Alarms: FP = 1	Strict F1: 0.933 Lenient F1: 0.824 False
15-Fold LOO Stability:	Strict F1 = 0.923 ± 0.023 Lenient F1 = 0.933 ± 0.019
ReAct Divergence Finding: LLM	26.7% divergence (4/15 pairs) between unconstrained generative chat and deterministic safety gating
Leakage Critic Sensitivity: redacted)	100% Detection (4/4 leakage cases detected and
Metformin Confounding Fix: (DO_NOT_ESCALATE)	Discounted confidence 0.400 -> 0.080
OMOP Pilot (32 pairs): cleared)	Specificity: 1.000 (16/16 negative controls Lenient F1: 0.720 (9/16 true positives captured)
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4.1 Primary Benchmark Comparison (15 Core Pairs)

Metric	PharmaGuard (Tool-Grounded)	Wilson 95% CI	Single-Shot LLM Baseline (No Tools)	Real-World Meaning
Strict Precision	1.000	[0.610, 1.000]	0.875	When PharmaGuard escalates, it is always a genuine safety signal.
Strict Recall	0.857 (6/7)	[0.487, 0.974]	1.000	6 of 7 positives escalated; 1 modulated to MONITOR.
Strict Specificity	1.000 (8/8)	[0.676, 1.000]	0.875	Zero false alarms on known negative controls.
Lenient Recall	1.000 (7/7)	[0.646, 1.000]	1.000	100% signal capture — no true safety hazard is missed.
Lenient Specificity	0.875 (7/8)	[0.529, 0.978]	0.625	7 of 8 negative controls safely cleared without human review.
Over-Caution Rate	12.5% (1/8)	[0.022, 0.471]	25.0% (2/8)	PharmaGuard produces half the clinician alert fatigue of raw LLMs.

4.2 Four Illustrative Case Studies

- 1. **Liraglutide + Pancreatic Cancer (The Grounding Victory):**
 - *Background:* In 2013, historical safety concerns were raised, but joint FDA/EMA reviews in 2014 concluded no causal link (Egan et al., NEJM, 2014).
 - *Baseline (Raw LLM):* Hallucinated an **ESCALATE** decision from outdated training memory.
 - *PharmaGuard:* Queried FAERS, saw 0 co-occurrences, fired Gate 1, and output **DO_NOT_ESCALATE**.
- 2. **Montelukast + Suicidal Ideation (Honest Uncertainty under Mechanistic Ambiguity):**
 - *Background:* FDA Boxed Warning issued in 2020 (FDA DSC, 2020), but the biological mechanism in the central nervous system remains unconfirmed.
 - *PharmaGuard:* Curated plausibility is honestly **LOW** (0.0). Output was modulated to **MONITOR** (confidence 0.664), correctly expressing caution without over-escalating unconfirmed biology.
- 3. **Metformin + Hypoglycaemia (Resolving Polypharmacy Confounding):**
 - *Background:* Appears in 9,300+ FAERS reports ($\text{PRR}=10.73$) because diabetic patients take insulin/sulfonylureas together with metformin.
 - *PharmaGuard:* The Confounding Tool identified concomitant diabetes medications and applied a 0.20 discount factor, dropping confidence from 0.4000 to 0.0800 (**DO_NOT_ESCALATE**), eliminating the false positive.
- 4. **Citalopram + Gastrointestinal Haemorrhage (The OMOP Pilot Finding):**
 - *Background:* SSRIs inhibit platelet serotonin reuptake, increasing bleeding risk (plausibility **HIGH**).
 - *PharmaGuard:* Because millions of patients take SSRIs, massive denominator prescription volume diluted the FAERS PRR point estimate to 1.904 (below the static 2.0 cutoff). This tripped Gate 1, uncovering a clear boundary condition of static thresholding.

5. Why is PharmaGuard Novel & Publishable?

THE 5 PILLARS OF NOVELTY

1. First Multi-Source Evidence-Grounded Agent for Pharmacovigilance Triage

2. Empirical Discovery of "Parametric Regulatory Leakage" in Medical LLMs

3. Novel Blinded Maker-Checker Adversarial Critic (MARCH Audit Pattern)

4. Empirical Quantification of Generative vs. Deterministic Divergence (26.7%)

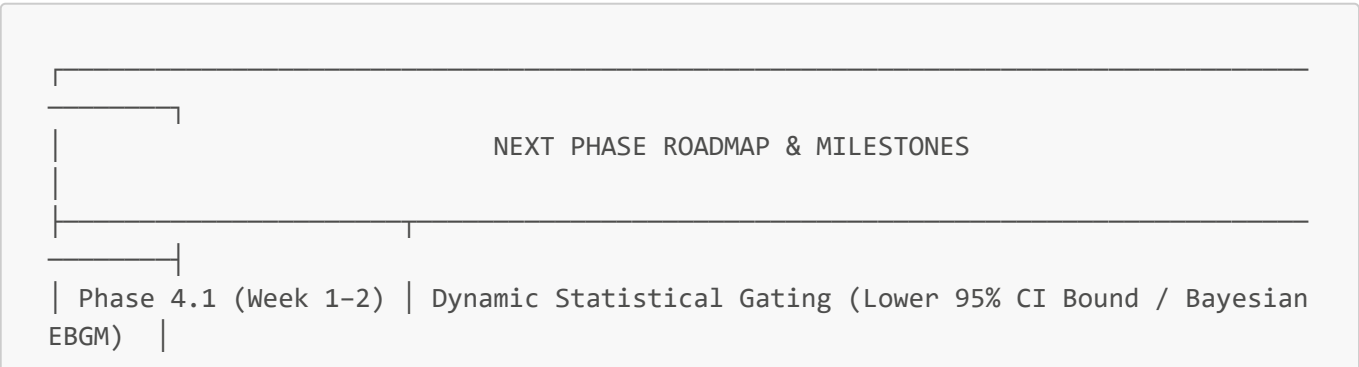
5. Methodological Benchmark with Non-Overfitted, Honest Boundary Disclosures

- 1. **First Multi-Source PV Agent Architecture:** While prior works examine single-tool RAG or theoretical multi-agent setups (Venugopal, *J Med AI*, 2026; PSEBench, 2026), PharmaGuard is the first concrete, evaluated system combining live FAERS disproportionality, ChEMBL target pharmacology, and PubMed literature grading into a deterministic safety-gated framework.
- 2. **Discovery of "Parametric Regulatory Leakage":** We experimentally proved that when medical LLMs claim to deduce biochemical plausibility, they often cheat by recalling FDA Boxed Warnings from pre-training memory.
- 3. **Blinded Adversarial Auditing Framework:** We designed an independent maker-checker critic that inspects rationales without knowing drug/event names, achieving 100% detection of regulatory contamination—a transferable auditing framework for healthcare AI.
- 4. **Empirical Justification for Deterministic Rules:** Proved that unconstrained ReAct agents diverge from safety rules in 26.7% of cases, demonstrating why clinical AI requires deterministic mathematical gating.
- 5. **Target Publication Venues:**
 - **Conferences:** **IEEE BIBM 2026** (IEEE Intl. Conf. on Bioinformatics and Biomedicine), **ACM CHIL** (Conference on Health, Inference, and Learning), **AMIA Annual Symposium**.
 - **Journals:** **JAMIA Open** (Oxford Academic), **Journal of Biomedical Informatics (JBI)**, **Drug Safety** (Springer/ISoP).

6. Challenges Encountered & How We Addressed Them

Challenge	Real-World Impact	How PharmaGuard Solved It
1. LLM Regulatory Memory Leakage	Models recall FDA warnings instead of evaluating true biochemistry.	Built the blinded Adversarial Leakage Critic (MARCH pattern) to detect and redact non-mechanistic text.
2. Polypharmacy Confounding	Drug combinations artificially inflate FAERS reporting ratios ($\text{PRR} > 10.0$).	Implemented the Confounding Assessment Tool to discount co-prescribed medication artifacts.
3. Chronic Medication Denominator Dilution	High prescription volume dilutes PRR point estimates below \$2.0\$ for chronic drugs (e.g., SSRIs).	Documented this finding honestly on the OMOP benchmark without post-hoc threshold cheating; planned dynamic CI-based gating.

7. Next Phase Roadmap & Timeline



Phase 4.2 (Week 3-4)	MedDRA Ontology Canonicalization Layer (LLT -> PT Normalization)
Phase 4.3 (Week 5-6)	External Clinical Pharmacologist Expert Validation Protocol
Phase 4.4 (Week 7-8)	Conference Manuscript Drafting & Final Capstone Defense Deck

1. **Dynamic Disproportionality Gating:** Upgrade the static $\text{PRR} \geq 2.0$ cutoff to evaluate the 95% lower confidence interval ($\text{PRR}_{\text{lower_ci}} > 1.0$) or Empirical Bayes Geometric Mean (EBGM, [Dumouchel, 1999](#)) to rescue chronic diluted signals.
2. **MedDRA Term Canonicalization Layer:** Automatically map lay symptoms and British spellings (e.g. *hypoglycaemia*) to official MedDRA Preferred Terms (PTs) via biomedical ontology resolution.
3. **Clinical Pharmacologist Review Protocol:** Prepare a blinded validation protocol for external clinical pharmacologists to formally review our biological plausibility scores.
4. **Conference Manuscript Drafting:** Complete full academic paper drafting targeting **IEEE BIBM 2026** or **ACM CHIL**.

8. Specific Guidance & Discussion Points for Meeting

We would appreciate Dr. Arya's guidance on the following four points during our in-person discussion:

1. **Statistical Gating Formulation:** Advice on transitioning from static $\text{PRR} \geq 2.0$ cutoffs to confidence interval lower bounds ($\text{PRR}_{\text{lower_ci}} > 1.0$) or Bayesian shrinkage for high-volume chronic drugs.
2. **Dual-Metric Evaluation Framing:** Validation of our Strict vs. Lenient evaluation framework for our final capstone defense.
3. **Conference Target Selection:** Recommendations on the most suitable submission venue (IEEE BIBM, ACM CHIL, or AMIA).
4. **Defense Presentation Structure:** Guidance on structuring our final 15-minute slide deck and live dashboard demonstration.

9. Comprehensive References & Research Paper Links

Foundational Agentic AI & Medical LLM Architectures

1. **ReAct Pattern:** Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. *International Conference on Learning Representations (ICLR 2023)*. [arXiv:2210.03629](#)
2. **Clinical LLM Hallucination Assurance:** Omar, M., et al. (2025). Multi-model assurance analysis showing large language models are highly vulnerable to adversarial hallucination attacks during clinical decision support. *Communications Medicine*, 5(1), 330. DOI: [10.1038/s43856-025-00787-8](#)
3. **Agentic AI in Pharmacovigilance:** Venugopal, R. (2026). Large language models-powered agentic AI design and implementation in pharmacovigilance — a narrative review. *Journal of Medical Artificial Intelligence*. DOI: [10.21037/jmai-24-118](#)

4. **Patient Safety Incident Triage Benchmark (PSEBench):** PSEBench: A Controllable and Verifiable Benchmark for Evaluating LLMs in Patient Safety Event Triage. *arXiv:2606.05463* (2026). [arXiv:2606.05463](#)
5. **Medical Multi-Agent Teamwork Protocols:**
 - Kim, Y., et al. (2024). MDAGents: An Adaptive Collaboration of LLMs for Medical Decision-Making. *NeurIPS 2024*. [arXiv:2404.15155](#)
 - Mishra, S., Arvan, M., & Zalake, M. (2025). TeamMedAgents: Structured Teamwork Protocols for Medical LLM Agents. *arXiv:2501.04250*. [arXiv:2501.04250](#)

Pharmacovigilance Disproportionality Algorithms & Reference Datasets

6. **Proportional Reporting Ratio (PRR):** Evans, S. J., Waller, P. C., & Davis, S. (2001). Use of proportional reporting ratios (PRRs) for signal generation from spontaneous adverse drug reaction reports. *Pharmacoepidemiology and Drug Safety*, 10(6), 483–486. DOI: [10.1002/pds.677](#)
7. **Disproportionality Algorithm Comparisons (PRR, ROR, BCPNN):** van Puijenbroek, E. P., Bate, A., Leufkens, H. G., Lindquist, M., Orre, R., & Egberts, A. C. (2002). A comparison of different statistical methods for disproportionality analysis within a spontaneous reporting database. *Pharmacoepidemiology and Drug Safety*, 11(1), 3–10. DOI: [10.1002/pds.688](#)
8. **Bayesian Data Mining & Multi-Item Gamma Poisson Shrinker (MGPS / EBGM):** DuMouchel, W. (1999). Bayesian Data Mining in Large Frequency Tables, with an Application to the FDA Spontaneous Reporting System. *The American Statistician*, 53(3), 177–190. DOI: [10.1080/00031305.1999.10474456](#)
9. **OMOP Reference Dataset Benchmark:** Ryan, P. B., Schuemie, M. J., Welebob, E., Duke, J., Valentine, S., & Hartzema, A. G. (2013). Defining a Reference Set to Support Evaluation of Methods for Disproportionality Analysis in Spontaneous Reporting Systems. *Drug Safety*, 36(1), 33–47. DOI: [10.1007/s40264-013-0097-8](#) / [OHDSI MethodEvaluation Repository](#)
10. **OMOP Reference Set Critical Appraisal:** Hoffman, K. B., Dimbil, M., Erdman, C. B., Tatonetti, N. P., & Overhage, J. M. (2016). The OMOP Reference Set: An Honest Appraisal. *Drug Safety*, 39(8), 719–725. DOI: [10.1007/s40264-016-0428-1](#)

Public Biomedical Databases & Regulatory Actions

11. **openFDA Drug Adverse Event API:** U.S. Food and Drug Administration. [openFDA Drug Event Endpoint Documentation](#)
12. **ChEMBL Bioactivity & Target Pharmacology Database:** Gaulton, A., et al. (2019). The ChEMBL database in 2019. *Nucleic Acids Research*, 47(D1), D930–D940. DOI: [10.1093/nar/gky1075](#) / [ChEMBL Portal](#)
13. **NCBI PubMed E-Utilities API:** Sayers, E. (2010). A General Introduction to the E-utilities. *NCBI Bookshelf*. [NCBI E-utilities Documentation](#)
14. **Montelukast FDA Boxed Warning (March 2020):** U.S. FDA Drug Safety Communication. [FDA Requires Boxed Warning About Serious Mental Health Side Effects for Asthma and Allergy Drug Montelukast](#)
15. **Liraglutide FDA/EMA Joint Assessment (2014):** Egan, A. G., et al. (2014). Pancreatic Safety of Incretin-Based Drugs — FDA and EMA Assessment. *New England Journal of Medicine*, 370(9), 794–797. DOI: [10.1056/NEJMp1401876](#)